Data Privacy: FinalSolutions

1. **Exact Score and Denoiser of a Two-Point Data Distribution** *(30 points)* Let the data be the fair two-point distribution $X_0 \in \{+1, -1\}$, $\Pr[X_0 = +1] = \Pr[X_0 = -1] = \frac{1}{2}$. Fix one diffusion noise level and form the forward sample $Y = a\,X_0 + \sqrt{v}\,\varepsilon$ with $\varepsilon \sim \mathcal{N}(0,1)$ independent of X_0, where $a = \sqrt{\bar{\alpha}} \in (0,1)$ and $v = 1 - \bar{\alpha}$. Throughout, write the Gaussian density, the logistic sigmoid, and the hyperbolic tangent as

$$f(x;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad \sigma(t) = \frac{1}{1+e^{-t}}, \qquad \tanh(t) = \frac{e^t - e^{-t}}{e^t + e^{-t}}.$$

   (a) *(5 points)* Show that the marginal density of Y is the two-component mixture $p(y) = \frac{1}{2}\,f(y;a,v) + \frac{1}{2}\,f(y;-a,v)$.

   (b) *(5 points)* Let $r_+(y) = \Pr[X_0 = +1 \mid Y = y]$. Show $r_+(y) = \sigma\big(\frac{2a}{v}y\big)$, and hence the optimal (posterior-mean) denoiser is $\mathbb{E}[X_0 \mid Y = y] = \tanh\big(\frac{a}{v}\,y\big)$. *(Hint: $r_+/r_- = f(y;a,v)/f(y;-a,v) = e^{2ay/v}$; then $\mathbb{E}[X_0 \mid y] = r_+ - r_-$.)*

   (c) *(10 points)* Show the exact score is $\partial_y \log p(y) = -\frac{y}{v} + \frac{a}{v}\tanh\big(\frac{a}{v}\,y\big)$, and verify consistency with (b) via the denoiser–score identity $\mathbb{E}[X_0 \mid Y = y] = \frac{1}{a}\,(y + v\,\partial_y \log p(y))$ (which you should also prove).

   (d) *(10 points)* Show separately that the denoiser $\mathbb{E}[X_0 \mid Y = y]$ converges to $\mathrm{sign}(y)$ as $\bar{\alpha} \to 1$, and converges to 0 as $\bar{\alpha} \to 0$. Interpret each limit.

Solution: Two-Point Diffusion Score.

(a) Conditioning on the bit, $Y \mid (X_0 = +1) \sim \mathcal{N}(a,v)$ and $Y \mid (X_0 = -1) \sim \mathcal{N}(-a,v)$, each with prior weight $\frac{1}{2}$. By the law of total probability,

$$p(y) = \tfrac{1}{2}\,f(y;a,v) + \tfrac{1}{2}\,f(y;-a,v).$$

(b) By Bayes' rule the two posteriors share the $\frac{1}{2}$ prior and the normalizer $p(y)$, so

$$\frac{r_+(y)}{r_-(y)} = \frac{f(y;a,v)}{f(y;-a,v)} = \exp\left(\frac{-(y-a)^2 + (y+a)^2}{2v}\right) = \exp\left(\frac{2ay}{v}\right).$$

Since $r_+ + r_- = 1$, $r_+(y) = \frac{e^{2ay/v}}{1+e^{2ay/v}} = \sigma\big(\frac{2a}{v}y\big)$ and $r_-(y) = \sigma\big(-\frac{2a}{v}y\big)$. The posterior mean is

$$\mathbb{E}[X_0 \mid Y = y] = (+1)\,r_+ + (-1)\,r_- = \sigma\big(\tfrac{2a}{v}y\big) - \sigma\big(-\tfrac{2a}{v}y\big) = \tanh\left(\frac{a}{v}\,y\right),$$

using $\sigma(2z) - \sigma(-2z) = \frac{e^{2z}-1}{e^{2z}+1} = \tanh z$ with $z = ay/v$.

(c) Differentiating the mixture,

$$p'(y) = \tfrac{1}{2}\,f(y;a,v)\big(-\tfrac{y-a}{v}\big) + \tfrac{1}{2}\,f(y;-a,v)\big(-\tfrac{y+a}{v}\big).$$

Dividing by $p(y)$ and recognizing $r_\pm$,

$$\begin{aligned}\partial_y \log p(y) &= -\tfrac{1}{v}\left[r_+(y-a) + r_-(y+a)\right] = -\tfrac{1}{v}\left[y - a(r_+ - r_-)\right] \\ &= -\frac{y}{v} + \frac{a}{v}\tanh\Big(\frac{a}{v}y\Big).\end{aligned}$$

Identity. For $Y = aX_0 + \sqrt{v}\,\varepsilon$ with Gaussian noise, $p(y) = \int p_{X_0}(x)\, f(y; ax, v)\, dx$, so

$$\begin{aligned}\partial_y p(y) &= \int p_{X_0}(x)\big(-\tfrac{y-ax}{v}\big)\, f(y; ax, v)\, dx \\ &= -\tfrac{y}{v}p(y) + \tfrac{a}{v}\, p(y)\, \mathbb{E}[X_0 \mid Y = y].\end{aligned}$$

Dividing by $p(y)$ gives $\partial_y \log p = -\frac{y}{v} + \frac{a}{v}\mathbb{E}[X_0 \mid y]$, i.e. $\mathbb{E}[X_0 \mid y] = \frac{1}{a}(y + v\,\partial_y \log p)$. Substituting the score above reproduces $\tanh(ay/v)$, matching (b).

(d) Recall $a = \sqrt{\bar\alpha}$, $v = 1 - \bar\alpha$, so $a/v = \sqrt{\bar\alpha}/(1-\bar\alpha)$.

Vanishing noise $\bar\alpha \to 1$: $a \to 1$, $v \to 0^+$, so $a/v \to +\infty$ and for every $y \neq 0$, $\tanh(ay/v) \to \operatorname{sign}(y)$. The denoiser snaps to the nearest clean value ± 1 — the bit is recoverable because almost no noise was added.

Saturating noise $\bar\alpha \to 0$: $a \to 0$, so $\tanh(ay/v) \to 0$ and $\mathbb{E}[X_0 \mid y] \to 0 = \mathbb{E}[X_0]$: the observation carries no information, so the best estimate is the prior mean. Correspondingly the score collapses to $-y/v$, exactly the score of the pure-noise Gaussian $\mathcal{N}(0, v)$.

2. **LiRA Closed Form and Decision Boundary** *(30 points)*
For a target record z, the per-shadow-model loss is Gaussian under the two membership hypotheses:

$$\ell \mid (z \in D) \sim \mathcal{N}(\mu_{\text{in}}, \sigma_{\text{in}}^2), \qquad \ell \mid (z \notin D) \sim \mathcal{N}(\mu_{\text{out}}, \sigma_{\text{out}}^2).$$

(a) *(10 points)* (*Warm-up.*) Derive the log-likelihood ratio

$$\log \Lambda(\ell) = \frac{(\ell - \mu_{\text{out}})^2}{2\sigma_{\text{out}}^2} - \frac{(\ell - \mu_{\text{in}})^2}{2\sigma_{\text{in}}^2} + \log\frac{\sigma_{\text{out}}}{\sigma_{\text{in}}}.$$

(Hint: subtract the two log-densities $\log \mathcal{N}(\ell; \mu, \sigma^2) = -\frac{(\ell-\mu)^2}{2\sigma^2} - \frac{1}{2}\log(2\pi\sigma^2)$.*)*

(b) *(10 points)* For $\sigma_{\text{in}} \neq \sigma_{\text{out}}$, show the region $\{\log\Lambda > \log\tau\}$ is a bounded interval, its complement, a half-line, all of $\mathbb{R}$, or empty, and say which from the sign of $\sigma_{\text{in}} - \sigma_{\text{out}}$. For equal variances $\sigma = \sigma_{\text{in}} = \sigma_{\text{out}}$ (and $\mu_{\text{in}} < \mu_{\text{out}}$), show it becomes a threshold $\mathcal{A}(z) = \mathbf{1}[\ell \leq c]$ and find c.
(Hint: $\log\Lambda$ *is a quadratic* $A\ell^2 + B\ell + C'$ *with* $A = \frac{1}{2}(\sigma_{\text{out}}^{-2} - \sigma_{\text{in}}^{-2})$*; the sign of* A *sets the shape.)*

(c) *(10 points)* With equal variances, derive the ROC: at false-positive rate α,

$$\text{TPR}(\alpha) = \Phi\left(\Phi^{-1}(\alpha) + \frac{\mu_{\text{out}} - \mu_{\text{in}}}{\sigma}\right),$$

where $\Phi(t) = \int_{-\infty}^{t} \frac{1}{\sqrt{2\pi}} e^{-s^2/2}\, ds$ is the standard normal CDF.
(Hint: solve $\text{FPR}(c) = \Phi(\frac{c-\mu_{\text{out}}}{\sigma}) = \alpha$ *for* c*, then substitute into* $\text{TPR}(c) = \Phi(\frac{c-\mu_{\text{in}}}{\sigma})$.*)*

Solution: LiRA Closed Form and Decision Boundary.

(a) Using $\log \mathcal{N}(\ell; \mu, \sigma^2) = -\frac{(\ell-\mu)^2}{2\sigma^2} - \frac{1}{2}\log(2\pi\sigma^2)$ and subtracting the out-density log from the in-density log, the $-\frac{1}{2}\log(2\pi)$ terms cancel and

$$\log \Lambda(\ell) = -\frac{(\ell - \mu_{\text{in}})^2}{2\sigma_{\text{in}}^2} + \frac{(\ell - \mu_{\text{out}})^2}{2\sigma_{\text{out}}^2} + \tfrac{1}{2}\log\frac{\sigma_{\text{out}}^2}{\sigma_{\text{in}}^2},$$

which is the claimed expression.

(b) Collecting powers of ℓ, $\log \Lambda(\ell) = A\ell^2 + B\ell + C'$ with $A = \frac{1}{2}(\sigma_{\text{out}}^{-2} - \sigma_{\text{in}}^{-2})$. Note $A < 0 \iff \sigma_{\text{in}} < \sigma_{\text{out}}$, so $\text{sign}(A) = \text{sign}(\sigma_{\text{in}} - \sigma_{\text{out}})$. The region $\{A\ell^2 + B\ell + (C' - \log\tau) > 0\}$ is:

- $\sigma_{\text{in}} < \sigma_{\text{out}}$ ($A < 0$, downward parabola): a *bounded interval* (c_-, c_+), degenerating to *empty* if the maximum is below $\log\tau$;
- $\sigma_{\text{in}} > \sigma_{\text{out}}$ ($A > 0$, upward parabola): the *complement of a bounded interval* $(-\infty, c_-) \cup (c_+, \infty)$, degenerating to *all of* $\mathbb{R}$ if the minimum exceeds $\log\tau$;
- $\sigma_{\text{in}} = \sigma_{\text{out}}$ ($A = 0$, affine): a *half-line.*

Equal variances. With $\sigma_{\text{in}} = \sigma_{\text{out}} = \sigma$ the quadratic terms cancel:

$$\log \Lambda(\ell) = \frac{(\ell - \mu_{\text{out}})^2 - (\ell - \mu_{\text{in}})^2}{2\sigma^2} = \frac{(\mu_{\text{out}} - \mu_{\text{in}})(\mu_{\text{in}} + \mu_{\text{out}} - 2\ell)}{2\sigma^2}.$$

Since $\mu_{\text{out}} > \mu_{\text{in}}$, $\log\Lambda$ is strictly decreasing in ℓ, so $\log\Lambda(\ell) \geq \log\tau \iff \ell \leq c$ with

$$c = \frac{\mu_{\text{in}} + \mu_{\text{out}}}{2} - \frac{\sigma^2 \log\tau}{\mu_{\text{out}} - \mu_{\text{in}}},$$

i.e. $\mathcal{A}(z) = \mathbf{1}[\ell \leq c]$ (low loss $\Rightarrow$ member).

(c) For $\mathcal{A} = \mathbf{1}[\ell \leq c]$, $\text{FPR}(c) = \Pr_{\text{out}}[\ell \leq c] = \Phi(\frac{c - \mu_{\text{out}}}{\sigma})$. Setting $\text{FPR} = \alpha$ gives $c = \mu_{\text{out}} + \sigma\,\Phi^{-1}(\alpha)$. Then

$$\text{TPR}(\alpha) = \Pr_{\text{in}}[\ell \leq c] = \Phi\left(\frac{c - \mu_{\text{in}}}{\sigma}\right) = \Phi\left(\Phi^{-1}(\alpha) + \frac{\mu_{\text{out}} - \mu_{\text{in}}}{\sigma}\right).$$

The ROC is governed by the standardized gap $(\mu_{\text{out}} - \mu_{\text{in}})/\sigma$.

3. **Why NPO's Gradient Stays Bounded and GA's Does Not** *(40 points)*
For LLM unlearning let $\pi_\theta(y \mid x) \in (0, 1]$ be smooth in θ, with $\pi_{\text{ref}}(y \mid x) \in (0, 1]$ a fixed reference. On the forget set $\mathcal{D}_F$ the two losses are

$$\mathcal{L}_{\text{GA}}(\theta) = \mathbb{E}_{\mathcal{D}_F}[\log \pi_\theta(y \mid x)], \qquad \mathcal{L}_{\text{NPO}}(\theta) = \frac{2}{\beta}\,\mathbb{E}_{\mathcal{D}_F}\left[\log\left(1 + \left(\tfrac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}\right)^\beta\right)\right] \quad (\beta > 0),$$

where $\sigma(t) = 1/(1 + e^{-t})$ is the logistic sigmoid. Assume $\|\nabla_\theta \pi_\theta(y \mid x)\| \leq G$, and differentiate sums and expectations term by term freely.

(a) *(10 points)* Let $u(\theta) = \beta \log(\pi_\theta(y \mid x)/\pi_{\text{ref}}(y \mid x))$. Show that

$$\mathcal{L}_{\text{NPO}}(\theta) = \tfrac{2}{\beta}\,\mathbb{E}[\log(1 + e^u)], \qquad \tfrac{\partial}{\partial u}\log(1 + e^u) = \sigma(u) \in (0, 1),$$

and that $\nabla_\theta u = \beta\,\nabla_\theta \log \pi_\theta$.
(Hint: $(\pi_\theta/\pi_{\text{ref}})^\beta = e^u$; $\frac{d}{du}\log(1 + e^u) = e^u/(1 + e^u) = \sigma(u)$, and $0 < \sigma < 1$ since $0 < e^u < 1 + e^u$; and π_{ref} is constant in θ.)

(b) *(10 points)* Using part (a) and the chain rule, show $\nabla_\theta \mathcal{L}_{\text{NPO}} = 2\,\mathbb{E}[\sigma(u)\,\nabla_\theta \log \pi_\theta]$, and conclude the uniform bound $\|\nabla_\theta \mathcal{L}_{\text{NPO}}\| \le 2\,\mathbb{E}[\|\nabla_\theta \log \pi_\theta\|]$.
(Hint: differentiate $\frac{2}{\beta}\log(1 + e^u)$ with $\nabla_\theta u = \beta \nabla_\theta \log \pi_\theta$; then use the triangle inequality $\|\mathbb{E}[V]\| \le \mathbb{E}[\|V\|]$ together with $\sigma(u) < 1$.)

(c) *(10 points)* Define the per-example NPO contribution $g_{\text{NPO}} := \sigma(u)\,\nabla_\theta \log \pi_\theta$. Show that

$$\|g_{\text{NPO}}\| = \sigma(u)\,\frac{\|\nabla_\theta \pi_\theta\|}{\pi_\theta}, \qquad \sigma(u) = \frac{(\pi_\theta/\pi_{\text{ref}})^\beta}{1 + (\pi_\theta/\pi_{\text{ref}})^\beta},$$

and hence that as $\pi_\theta(y \mid x) \to 0$, $\|g_{\text{NPO}}\| \le \frac{G}{\pi_{\text{ref}}^\beta}\,\pi_\theta^{\beta - 1}$.
(Hint: $\nabla_\theta \log \pi_\theta = \nabla_\theta \pi_\theta/\pi_\theta$; bound $1 + (\pi_\theta/\pi_{\text{ref}})^\beta \ge 1$ to get $\sigma(u) \le (\pi_\theta/\pi_{\text{ref}})^\beta$; combine with $\|\nabla_\theta \pi_\theta\| \le G$.)

(d) *(10 points)* Using part (c), show that as $\pi_\theta \to 0$ the NPO per-example gradient $\|g_{\text{NPO}}\|$ stays bounded for $\beta \ge 1$, while the GA per-example gradient $\|\nabla_\theta \log \pi_\theta\|$ diverges. Why does this make NPO stable but GA collapse?
(Hint: read off the limits of $\pi_\theta^{\beta-1}$ from (c); the GA norm is $\|\nabla_\theta \pi_\theta\|/\pi_\theta$, whose numerator stays bounded while $\pi_\theta \to 0$.)

Solution: Why NPO's Gradient Stays Bounded and GA's Does Not.

(a) Since $(\pi_\theta/\pi_{\text{ref}})^\beta = e^{\beta \log(\pi_\theta/\pi_{\text{ref}})} = e^u$, the loss is $\mathcal{L}_{\text{NPO}} = \frac{2}{\beta}\mathbb{E}[\log(1 + e^u)]$. Differentiating, $\frac{d}{du}\log(1 + e^u) = \frac{e^u}{1+e^u} = \sigma(u)$, and $0 < \sigma(u) < 1$ because $0 < e^u < 1 + e^u$ for all finite u. Finally π_{ref} does not depend on θ, so $u = \beta(\log \pi_\theta - \log \pi_{\text{ref}})$ gives $\nabla_\theta u = \beta\,\nabla_\theta \log \pi_\theta$.

(b) By the chain rule (differentiating term by term),

$$\nabla_\theta \mathcal{L}_{\text{NPO}} = \tfrac{2}{\beta}\,\mathbb{E}[\sigma(u)\,\nabla_\theta u] = \tfrac{2}{\beta}\,\mathbb{E}[\sigma(u)\,\beta\,\nabla_\theta \log \pi_\theta] = 2\,\mathbb{E}[\sigma(u)\,\nabla_\theta \log \pi_\theta].$$

Taking norms and using the triangle inequality $\|\mathbb{E}[V]\| \le \mathbb{E}[\|V\|]$ with $V = 2\sigma(u)\nabla_\theta \log \pi_\theta$ and $\sigma(u) < 1$,

$$\|\nabla_\theta \mathcal{L}_{\text{NPO}}\| \le 2\,\mathbb{E}[\sigma(u)\,\|\nabla_\theta \log \pi_\theta\|] \le 2\,\mathbb{E}[\|\nabla_\theta \log \pi_\theta\|].$$

This bound holds for every θ — it is uniform.

(c) Since $\nabla_\theta \log \pi_\theta = \nabla_\theta \pi_\theta/\pi_\theta$, the definition of g_{NPO} gives

$$\|g_{\text{NPO}}\| = \sigma(u)\,\|\nabla_\theta \log \pi_\theta\| = \sigma(u)\,\frac{\|\nabla_\theta \pi_\theta\|}{\pi_\theta}.$$

Writing $\rho := (\pi_\theta/\pi_{\text{ref}})^\beta = e^u$, we have $\sigma(u) = \rho/(1+\rho)$. As $\pi_\theta \to 0$, $\rho \to 0$; and $1 + \rho \geq 1$ gives $\sigma(u) \leq \rho$. With $\|\nabla_\theta \pi_\theta\| \leq G$,

$$\|g_{\text{NPO}}\| \leq \frac{(\pi_\theta/\pi_{\text{ref}})^\beta\, G}{\pi_\theta} = \frac{G}{\pi_{\text{ref}}^\beta}\, \pi_\theta^{\beta-1}.$$

(d) For $\beta > 1$, $\pi_\theta^{\beta-1} \to 0$ so $\|g_{\text{NPO}}\| \to 0$; for $\beta = 1$, $\|g_{\text{NPO}}\| \leq G/\pi_{\text{ref}}$, a finite constant. By contrast the GA per-example contribution has norm $\|\nabla_\theta \pi_\theta\|/\pi_\theta \sim 1/\pi_\theta \to \infty$ (the numerator stays bounded while $\pi_\theta \to 0$). As the forget likelihood $\pi_\theta \to 0$ the saturating factor $\sigma(u) \to 0$ at rate π_θ^β, which cancels (for $\beta \geq 1$) the $1/\pi_\theta$ growth of the log-likelihood gradient — that is why sigmoid saturation keeps the NPO step bounded and prevents the GA-style collapse.